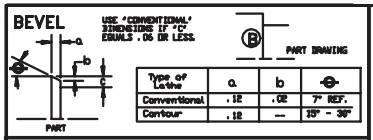
SHAFT STANDARDS

003425-000

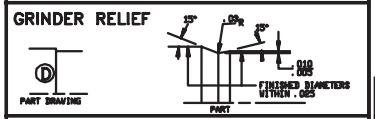
BEVEL

USE 'CONVENTIONAL' DIMENSIONS IF 'C' EQUALS .06 OR LESS.

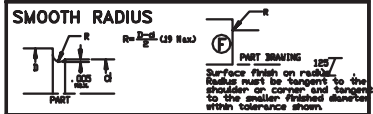


Type of Lathe	a	b	c
Conventional	.12	.02	7° REF.
Contour	.12	—	15° - 30°

GRINDER RELIEF

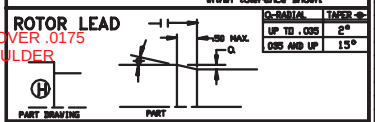


SMOOTH RADIUS



Surface finish on radius. Radius must be tangent to the shoulder or corner and tangent to the smaller finished diameter within tolerance shown.

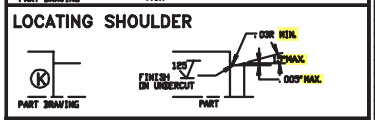
ROTOR LEAD



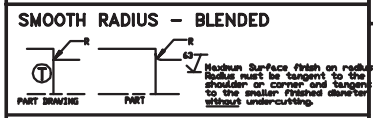
O-RADIAL	TAPER
UP TO .035	2°
.035 AND UP	15°

FOR 2°, A OVER .0175 HAVE SHOULDER

LOCATING SHOULDER

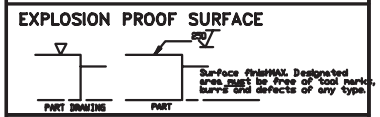


SMOOTH RADIUS - BLENDED



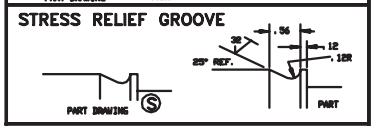
Maximum Surface finish on radius. Radius must be tangent to the shoulder or corner and tangent to the smaller finished diameter without undercutting.

EXPLOSION PROOF SURFACE



Surface finishMAX. Designated area must be free of tool marks, burrs and defects of any type.

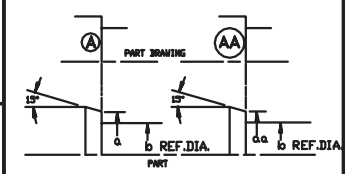
STRESS RELIEF GROOVE



SHOULDERS STRESS RELIEF

SEE SHEET 2 FOR ADDITIONAL SYMBOLS

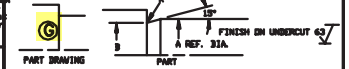
BEVEL - BEARING SHOULDER



BRG Size	b	a 200 SERIES (A)	aa 300 SERIES (AA)
03	.6695	.807/.787	.895/.875
04	.7878	.989/.959	1.036/1.016
05	.9847	1.192/1.172	1.240/1.220
06	1.1815	1.426/1.406	1.489/1.469
07	1.3785	1.634/1.614	1.708/1.688
08	1.5753	1.831/1.811	1.949/1.929
09	1.7722	2.028/2.008	2.146/2.126
10	1.9690	2.225/2.205	2.362/2.342
11	2.1660	2.461/2.441	2.579/2.559
12	2.3628	2.737/2.717	2.855/2.835
13	2.5597	2.933/2.913	3.051/3.031
14	2.7565	3.130/3.110	3.248/3.228
15	2.9534	3.327/3.307	3.445/3.425
16	3.1502	3.524/3.504	3.708/3.688
17	3.3472	3.760/3.740	3.918/3.898
18	3.5440	3.957/3.937	4.114/4.094
19	3.7409	4.233/4.213	4.311/4.291
20	3.9377	4.429/4.409	4.508/4.488
21	4.1348	4.626/4.606	4.705/4.685
22	4.3316	4.823/4.803	4.902/4.882
24	4.7253	5.217/5.197	5.296/5.276
26	5.1191	5.689/5.669	5.847/5.827
28	5.5128	6.083/6.063	6.240/6.220
30	5.9071	6.477/6.457	6.634/6.614
32	6.3008	6.870/6.850	7.028/7.008
34	6.6945	7.422/7.402	7.422/7.402

GRINDER RELIEF - BEARING SHOULDER

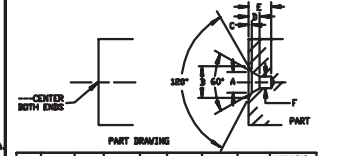
-.010 P/S



BRG Size	A	B	R
03	.6695	.653/.648	.031/.020
04	.7878	.772/.762	.040/.030
05	.9847	.969/.959	.040/.030
06	1.1815	1.166/1.156	.040/.030
07	1.3785	1.368/1.348	.040/.030
08	1.5753	1.565/1.545	.040/.030
09	1.7722	1.762/1.742	.040/.030
10	1.9690	1.959/1.939	.040/.030
11	2.1660	2.156/2.136	.060/.050
12	2.3628	2.353/2.333	.060/.050
13	2.5597	2.550/2.530	.060/.050
14	2.7565	2.746/2.726	.060/.050
15	2.9534	2.943/2.923	.060/.050
16	3.1502	3.140/3.120	.080/.050
17	3.3472	3.337/3.317	.080/.050
18	3.5440	3.534/3.514	.080/.050
19	3.7409	3.731/3.711	.080/.050
20	3.9377	3.928/3.908	.080/.050
21	4.1348	4.125/4.105	.080/.050
22	4.3316	4.322/4.302	.080/.050
24	4.7253	4.715/4.695	.080/.060
26	5.1191	5.109/5.089	.100/.090
28	5.5128	5.503/5.483	.100/.090
30	5.9071	5.897/5.877	.100/.090
32	6.3008	6.291/6.271	.100/.090
34	6.6945	6.684/6.664	.100/.090

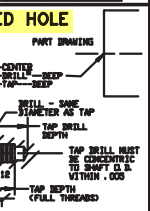
NOTE: FOR 'A' DIAMETERS NOT LISTED; B = A - .020 MAXIMUM

CENTERS



STANDARD	CENTER	A	B MAX.	C	D	E	F	MAX LOAD ON CENTER (LBS.)
CB	0 12	.150	.188	.061	.087	.149	.062	65
CC	0 13	.200	.250	.014	.106	.200	.094	135
CD	0 14	.250	.312	.018	.140	.250	.109	220
CE	0 15	.320	.438	.022	.190	.349	.125	420
CF	0 17	.500	.625	.026	.279	.499	.219	875
CG	0 18	.640	.750	.043	.346	.625	.250	1300
CH	0 20	.800	1.000	.039	.436	.801	.375	2500
CJ	1-1/4	1.000	1.250	.072	.613	.988	.375	3700
CK	1-7/8	1.500	1.875	.109	1.062	1.457	.375	9100
CL	2-1/4	1.875	2.250	.109	1.437	1.782	.375	13000
CM	1-1/16	.750	1.062	.090	.520	.782	.250	1875

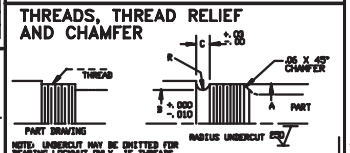
CENTER WITH TAPPED HOLE



CENTER	MAX. TAP	MIN. TAP
0 13	8-40	8-40
0 14	8-32	8-32
0 15	1/4-20	8/10-24
0 17	5/16-18	1/4-20
0 18	3/8-16	5/16-18
0 20	1/2-13	1/2-13
1-1/4	7/8-9	1/2-13
1-7/8	1"-8	1/2-13
2-1/4	1-1/8-7	1/2-13

*REFERENCE-TAPE DRILLS PER 3425-S, 70% THREAD

THREADS, THREAD RELIEF AND CHAMFER



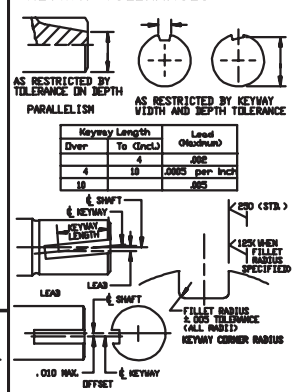
NOTE: UNDERCUT MAY BE OMITTED FOR BEARING LOCKNUT ONLY, IF THREADS ARE MILLED OR MACHINED ON NC EQUIP.

SEE 411179 FOR PITCH DIAMETERS

Table I Class 2A				
Thrd size A	Thrds Per In.	B	C	R
3/8 .375	16	.250	.09	.03
	24	.313	.09	.03
1/2 .500	13	.375	.09	.03
	20	.425	.09	.03
5/8 .625	11	.500	.12	.05
	18	.531	.12	.05
3/4 .750	10	.594	.12	.05
	16	.656	.12	.05
7/8 .875	9	.719	.12	.05
	14	.781	.12	.05
1" 1.000	8	.812	.16	.06
	14	.906	.16	.06
1 1/8 1.125	7	.938	.16	.06
	8	.938	.16	.06
	12	1.000	.16	.06
1 1/4 1.250	7	1.062	.16	.06
	8	1.062	.16	.06
	12	1.125	.16	.06
1 3/8 1.375	6	1.156	.19	.06
	8	1.188	.19	.06
	12	1.250	.19	.06
1 1/2 1.500	6	1.281	.19	.06
	8	1.312	.19	.06
	12	1.375	.19	.06
1 5/8 1.625	8	1.438	.19	.06
	5	1.500	.25	.08
1 3/4 1.750		1.500	.19	.06
1 7/8 1.875	8	1.688	.19	.06
	4.5	1.719	.25	.08
2" 2.000	8	1.844	.19	.06

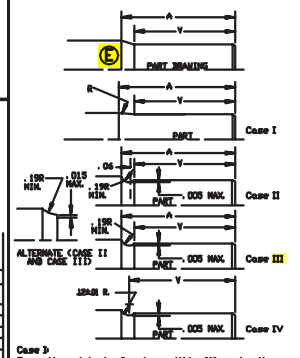
Table II Class 3A				
Brg Size	Thrd size A	B	C	R
03	.664-32	.616	.06	.02
04	.781-32	.730	.06	.02
05	.969-32	.921	.06	.02
06	1.172-18	1.095	.09	.03
07	1.376-18	1.298	.09	.03
08	1.563-18	1.485	.09	.03
09	1.767-18	1.689	.12	.05
10	1.967-18	1.889	.12	.05
11	2.157-18	2.079	.12	.05
12	2.360-18	2.282	.12	.05
13	2.548-18	2.470	.12	.05
14	2.751-18	2.673	.12	.05
15	2.938-18	2.861	.16	.06
16	3.137-18	3.025	.16	.06
17	3.340-18	3.228	.16	.06
18	3.537-18	3.415	.16	.06
19	3.730-18	3.618	.16	.06
20	3.918-18	3.806	.16	.06
21	4.128-18	4.010	.16	.06
22	4.325-18	4.213	.16	.06
24	4.716-18	4.604	.16	.06
26	5.106-18	4.994	.16	.06
28	5.497-18	5.385	.16	.06
30	5.888-18	5.776	.16	.06
32	6.284-8	6.120	.25	.08
34	6.659-8	6.495	.25	.08
36	7.034-8	6.906	.25	.08
38	7.409-8	7.281	.25	.08
40	7.847-8	7.669	.25	.08
44	8.659-8	8.464	.25	.08
48	9.442-8	9.227	.25	.08
52	10.198-8	9.977	.25	.08
56	11.004-8	10.789	.25	.08

KEYWAY TOLERANCES



Keyway Length		Lead
Over	To (incl.)	(Maximum)
	4	.002
4	10	.0005 per inch
10		.005

EXTENSION SHOULDERS



Case I:
Tag adjacent body diameters within .010 each other:
1. "V" may equal "A".
2. "R" is natural corner radius of grinding wheel.

Case II:
Shaft made on conventional lathe:
1. Use .19 Min radius on tool. Locate as shown.
2. "A" is maximum.
3. Surface finish on radius 500 MAX.

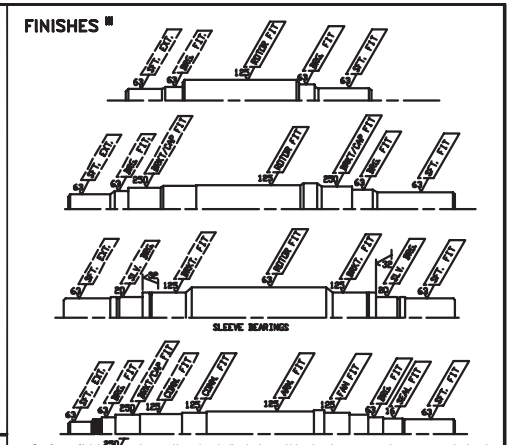
Case III:
Shaft made on contour NC lathe:
1. "R" should be such that the radius will be tangent to the shoulder or corner and also tangent to the finished O.D. to .005 undercut.
2. "V" is minimum.
3. Surface finish on radius 500 MAX.

Case IV:
FOR "PROP" PRACTICE
1. .12 R) should be tangent to finished O.D. to .005 undercut.
2. "V" is to center of .12 radius.
3. Surface finish on radius 125 MAX.

TOLERANCES UNLESS OTHERWISE SHOWN

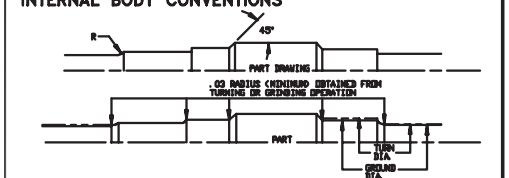
ONE PLACE DECIMAL	±. 1
TWO PLACE DECIMAL	±. 02
THREE PLACE DECIMAL	±. 003
FOUR PLACE DECIMAL	±. 0003

FINISHES [1]

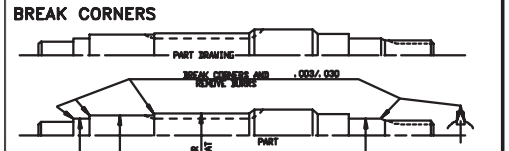


[1] Surface finish 250 unless otherwise indicated on this drawing or as shown on part drawing.

INTERNAL BODY CONVENTIONS



BREAK CORNERS



General Notes:
1 - Keyway Cutter Diameters
Widths up to .875 inclusive - "3" Dia. max.
Widths 1 and up - "4" Dia. max.
Larger dia. cutters may be required for special conditions.
2 - Maximum runout (T.I.R.)

SHAFT TOLERANCE	FRAME SIZE	MAX. RUNOUT
.001 MAX.	48 THRU 320	.001 TIR
	ALL J FRAMES	.001 TIR
	326 THRU 500	.0025 TIR
	OVER 500	.002 TIR
GREATER THAN .001	ALL FRAME SIZES	.003 TIR

3 - Out-of-Squareness of bearing shoulder .002 max.
4 - Out-of-Roundness of bearing seat .0002
5 - Unless shown otherwise on shaft part drawing, shaft standard 13425 shall be followed.
6 - RSN only - shaft chatter limit per 40B003-28
7 - RSN only - Taper on bearing seat 0.0002 max.

DR.	N. F. JUESCHKE 8/4/78	RSP 3-0096-01-751-604	
		DRAFTING MANUAL	3425
CK.	N. F. JUESCHKE	DISTRIBUTION- AG, AH, GG, MM, RC, MK, SN SD, ST ORIGINAL AT - RRC	
APP.	E. J. HINGER		SHEET 1 OF 2

C/R 245082, 276120, 306223, 314958, 315045, 325986, 335210 354492, 380817

003425-000

REV. DESC: REMOVE NOTE "1" FROM FINISHES SECTION OF STANDARD		
REV. LTR: A	VERSION: 01	TDR: 000000986548
FILE: \MGA\00001\347	REVISED: 01:38:21 06/28/2016	BY: MGHJMZ2
MTL: -		

BALDOR

SHAFT STANDARDS

SH 1 of 2